

How much can a black-hole image tell you about spin?

Amortized simulation-based inference of Kerr spin and inclination: status report

kerr-sbi project notes

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Abstract

We infer the spin a and inclination i of a Kerr black hole from noisy 64×64 ray-traced images of a thin accretion disk, using a CNN-embedded neural spline flow trained on simulations from the `nullgeo` ray tracer. A local Fisher analysis shows that spin information grows steeply with inclination, and a noise level $\sigma_n=20$ was chosen so that spin ranges from well constrained ($\sigma_a^{\text{CRB}} \approx 0.014$) to prior-dominated (≈ 0.27). A pilot network trained on 1,800 images in six CPU-minutes gains 1.95 nats per observation over the prior. It halves the spin and inclination errors and gives near-nominal global coverage (90% intervals cover 88.5% / 91.1%). Against an explicit-likelihood reference that passes prescribed finite numerical checks, its spin posteriors are typically $\sim 16\%$ too wide. The most important physical finding is that the spin signal comes largely from the assumption that the disk ends at the ISCO. With the inner edge fixed at $8M$, the image difference between $a=0$ and 0.98 shrinks by $7\text{--}23\times$. The exploratory free-radius pilot suggests weaker face-on spin constraints; one selected numerically checked observation demonstrates that an ISCO-assuming model can confidently exclude the true spin. These development results do not establish population calibration.

1 Problem and forward model

Each observation is produced by a deterministic simulator and a stochastic observation model. `nullgeo` (v0.2.0, used strictly as a black-box CLI) renders a Kerr hole ($M=1$) with a geometrically thin, co-rotating “stylized” disk. Its emission is $I \propto g^3(r/r_{\text{in}})^{-2}$, the disk extends from the ISCO to $15M$, and the camera sits at $R=85M$ with a 22° field of view and 9×9 supersampling (≈ 3.5 s per render). At the M1 gate it was chosen over a 40,000 K blackbody disk. It is exactly achromatic, so all information is in luminance, and its peak brightness varies a moderate $48\times$ across the parameter grid. Preprocessing takes linear luminance, applies a $\sigma=1$ px Gaussian PSF and normalizes the flux to a mean pixel value of 1, so absolute brightness is discarded. Independent Gaussian noise with $\sigma_n=20$ is then added. The prior is $a \sim \mathcal{U}[0, 0.98]$ and $\cos i \sim \mathcal{U}[\cos 80^\circ, \cos 5^\circ]$.

Fig. 1 shows the scale of the problem. The median peak pixel value on the original Fisher grid is 9.9, below $\sigma_n=20$. This is not a universal pixel bound: among 80 images inspected in the foundation audit, 30 contain a pixel at least as large as 20, and the maximum is 75.06. Whole-image information can remain substantial even when individual pixels have low signal-to-noise. Visually, the observations are almost pure noise.

2 Identifiability: the local Fisher map

Because the renderer is deterministic and the noise is Gaussian, the likelihood is explicit: $\log L(\theta) = -\|y - f(\theta)\|^2 / 2\sigma_n^2$. Fisher information $F = J^\top J / \sigma_n^2$ was computed on an 8×8 grid from central differences of f (328 renders plus 48 step-size, corner and supersampling checks; the prescribed checks at those ISCO-model locations and step sizes agreed within 12%). These finite checks do not certify continuum derivatives or derivatives in the fixed-radius model. At $\sigma_n=20$ (Fig. 2):

- σ_a^{CRB} spans 0.014–0.266 (median 0.124). The prior s.d. is 0.283, so the design covers regimes from sharply informative to barely constrained, which is what makes calibration testing meaningful.
- σ_i^{CRB} spans 2.3° – 15.5° . Both parameters are best measured edge-on, where beaming and the lensed arc carry the signal.
- Spin and inclination are almost locally orthogonal. The joint-to-conditional spin bound ratio is at most 1.16, so any degeneracy at this level is nonlinear or global rather than local.

3 Neural posterior estimation

Model. A 5-layer strided CNN with no global pooling (positional information matters) produces a 64-d embedding. This conditions a 6-layer masked-autoregressive rational-quadratic spline flow over logit-transformed,

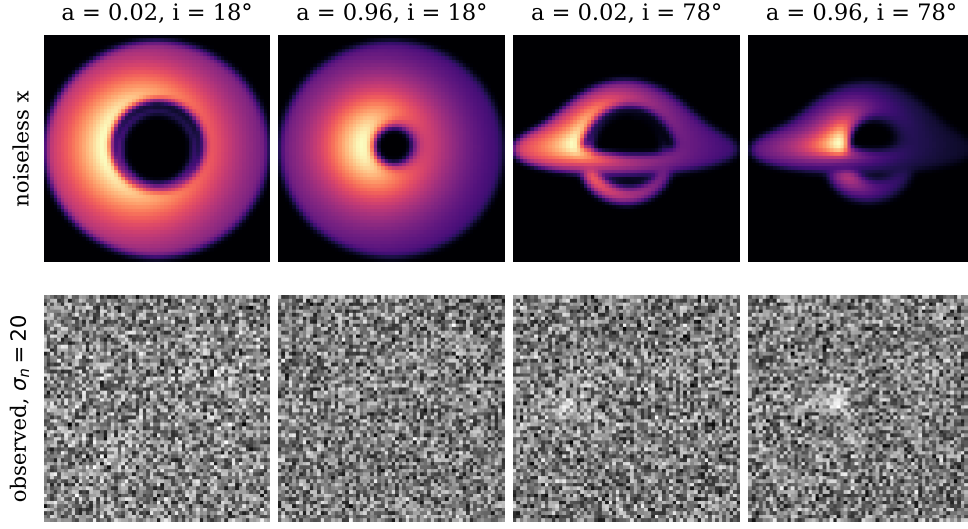


Figure 1: Top: preprocessed noiseless images (asinh stretch) at low/high spin and face-on/edge-on inclination. High spin moves the ISCO inward, shrinking the central hole; high inclination produces Doppler-beamed asymmetry (bright approaching side on the left) and a lensed far-side arc. Bottom: the same images with the adopted noise $\sigma_n=20$, which is what the network actually sees.

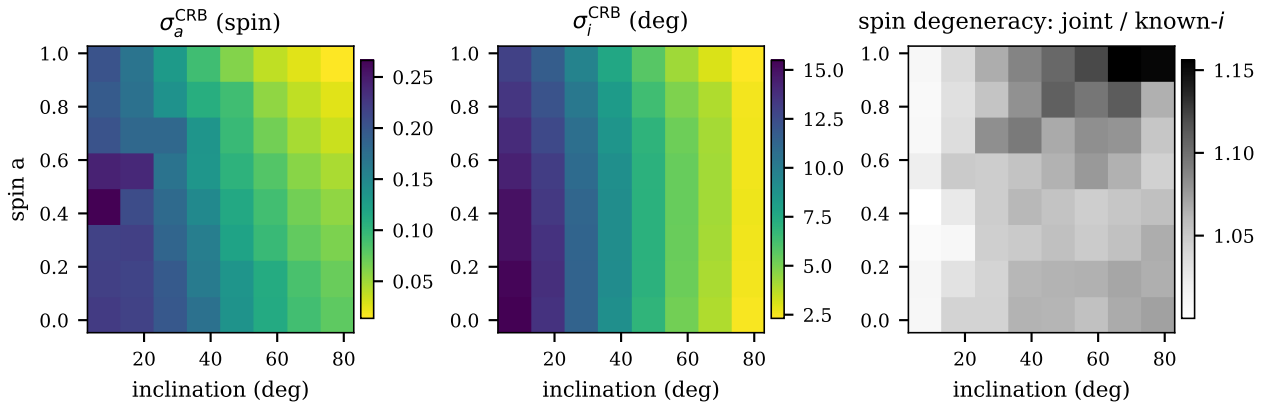


Figure 2: Local Cramér–Rao bounds at $\sigma_n=20$ (lighter = tighter). Left/centre: marginal bounds for spin and inclination. Right: ratio of the joint spin bound to the bound with inclination known; values near 1 mean weak local a – i degeneracy. These are finite-difference Fisher estimates, not guaranteed Bayesian widths.

standardized $(a, \cos i)$. The model has 966k parameters and is trained jointly in JAX/Equinox/FlowJAX with fresh noise in every batch. Inputs pass through the bijection $z = \text{asinh}(x_{\text{obs}}/s)$.

Pilot. Training used 2,000 prior-sampled renders (1,800 fitting and 200 validation) and took 365s on a laptop CPU, with early stopping at epoch 70 of 90. A noiseless 256-image overfit check reached NLL -2.59 against a prior NLL of 2.83. Evaluation used 1,000 independent development observations (seed 3, fixed noise) with 1,999 posterior draws each. The final seed-2 test set remains untouched.

	spin a	inclination i
RMSE, NPE posterior mean (prior-mean baseline)	0.128 (0.284)	7.37° (17.4°)
Coverage of 68 / 90 / 95% equal-tailed intervals	66.8 / 88.5 / 94.6%	69.8 / 91.1 / 95.5%
Spearman corr. of NPE s.d. with Fisher s.d. ($n=914$)	0.82	0.79

The mean NLL is 0.898 against a prior NLL of 2.848, a gain of 1.95 nats per observation. With image and parameter labels shuffled, the NLL rises to 7.27 (first 200 observations), confirming that the network uses the images. A data ladder of 450 / 900 / 1,800 fitting images gave gains of 1.78 / 1.93 / 1.95 nats. The paired improvement is 0.146 ± 0.022 nats from 450 to 900 images but only 0.020 ± 0.014 from 900 to 1,800. Returns therefore diminish at this architecture and budget, though these are single-seed runs with differing schedules.

Global coverage is close to nominal, but the regional picture is more informative (Fig. 3b). Spin RMSE falls from 0.21 face-on ($5-30^\circ$) to 0.068 edge-on ($65-80^\circ$), and 90% spin coverage face-on is only 80%. Conditional-on-truth coverage need not equal the nominal level for a correct Bayesian posterior, because prior shrinkage matters most exactly where the data are weak. This result is therefore a diagnostic target, not proof of miscalibration. The numerical likelihood references below test estimator agreement on selected development cases; they do not alone separate all causes of population or binwise coverage discrepancies.

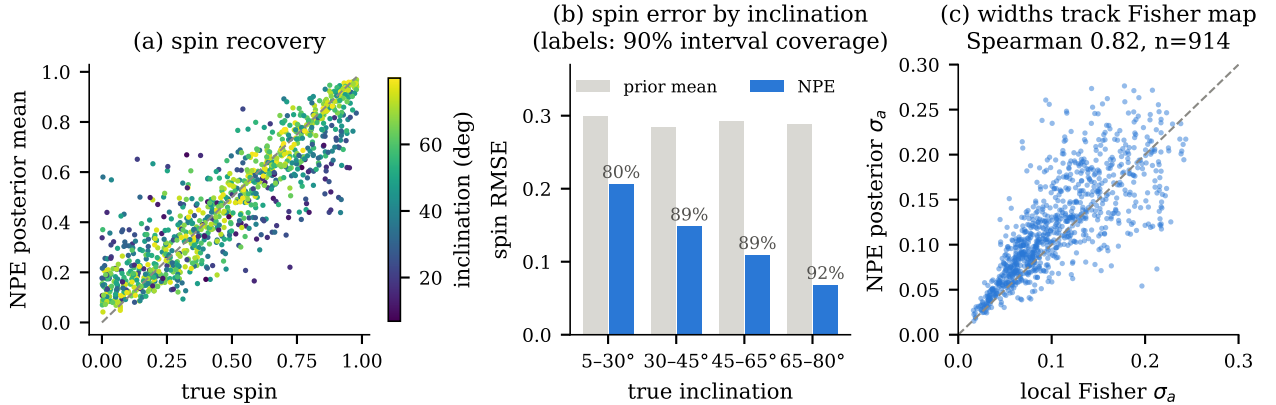


Figure 3: Pilot NPE on 1,000 development observations. (a) Posterior-mean spin vs. truth, coloured by inclination: edge-on systems (yellow) lie close to the diagonal; face-on systems shrink toward the prior mean. (b) Spin RMSE by inclination bin, with the 90%-interval coverage printed on each bar. (c) Posterior spin s.d. against the interpolated Fisher bound; NPE widths track the physics, with median ratio 1.16.

4 Numerical explicit-likelihood reference posteriors

The explicit Gaussian likelihood allows numerical reference posteriors without a neural network, provided the image manifold $f(a, \cos i)$ is interpolated accurately enough. Images were interpolated linearly on an adaptive Delaunay mesh of 2,385 rendered nodes, refined toward posterior mass. The Gaussian likelihood of each interpolated image was evaluated analytically, and its parameter-space integral was approximated using 512^2 midpoint quadrature. Six archetypes, each with three noise realizations, gave 18 cases. All 18 pass two consecutive mesh refinements (maximum marginal-CDF change 0.004) and both quadrature doublings. All 1,000/1,000 held-out development renders and 36/36 fresh posterior-region renders interpolate within 0.25 noise units (worst 0.11). The maximum log-likelihood error at the truths is 0.035. Passing these finite interpolation and integration checks does not establish mathematical exactness or continuum ray-tracing accuracy.

Compared with these numerically checked references (Fig. 4), the NPE reproduces the qualitative geometry: long prior-edge “bananas” face-on and compact ellipses edge-on. It is not quantitatively exact. The median marginal CDF discrepancy is 0.195 for spin and 0.138 for $\cos i$. The ratio of NPE to reference spin s.d. has a median of

1.16 (range 0.80–1.63). The median width excess is a selected-case approximation discrepancy. Widths alone do not establish conservative calibration, and the range includes narrower-than-reference results. Mean shifts and posterior shape also matter. The worst discrepancies occur in the narrowest, high-spin edge-on posteriors (e.g. case $a=0.95$, $i=74^\circ$), where small shifts are large relative to widths of ~ 0.02 .

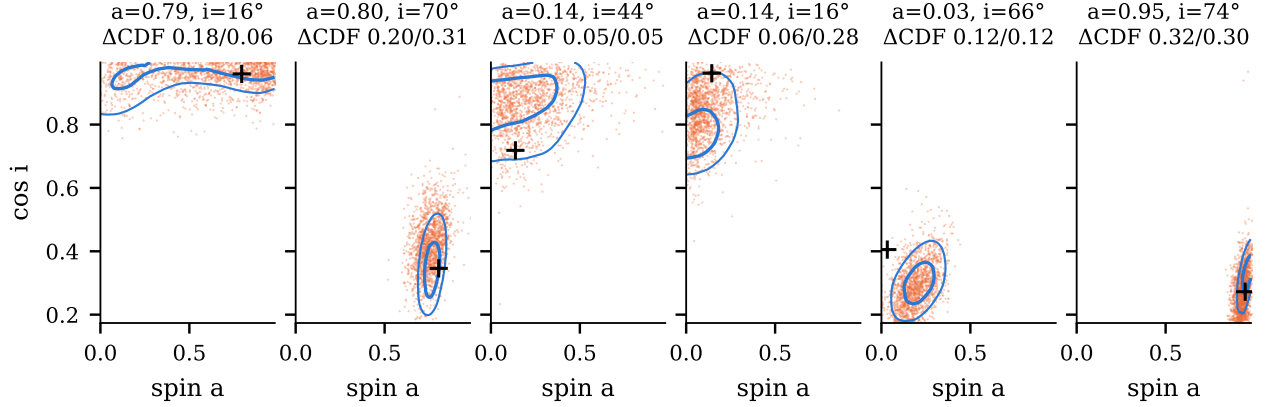


Figure 4: Numerically checked reference posteriors (blue: 50% and 90% highest-density contours) vs. NPE samples (orange) for the six archetypes (noise seed 104; + marks the truth). Panel titles give the marginal CDF discrepancy for spin / $\cos i$. Face-on posteriors are strongly non-Gaussian and prior-truncated, so Fisher widths cannot describe them.

5 How much of the spin signal is the ISCO assumption?

In this forward model, the disk’s inner edge sits exactly at $r_{\text{ISCO}}(a)$, which moves from $6M$ to $\approx 1.6M$ as $a \rightarrow 0.98$. This is also the key assumption behind real continuum-fitting spin measurements. Two experiments tested how much the spin constraint depends on it.

Known, fixed edge. Table 1 gives the noise-scaled full-image separation $D = \|f(0.98, i) - f(0, i)\| / \sigma_n$ at known inclination. Here $D^2/2$ is the expected log-likelihood ratio between the two hypotheses. Pinning the edge at $8M$, outside every ISCO in the prior, cuts the separation by $23\times$ face-on and $7\times$ edge-on (stable to $< 0.8\%$ under 15×15 supersampling). The residual edge-on signal presumably comes from spin-dependent lensing and Doppler structure at fixed radius (Fig. 5a).

The fixed-radius derivative follow-up tested four locations at half, original and double parameter steps with 9×9 and 15×15 sampling, plus 21×21 at the least stable location. At $a = 0.826$, $i = 18^\circ$, the 21×21 spin derivative changes by 56% when the step is halved and 46% when doubled, failing the retained 20% criterion. Tightening integrator tolerance separately has negligible effect in the tested case. Fixed-radius Fisher claims therefore remain unresolved; finite endpoint image separation is the supported comparison, and a three-parameter Fisher map is deferred.

Inclination	18°	45°	68°	78°
Disk at ISCO	3.21	5.66	11.01	14.63
Disk edge fixed at $8M$	0.14	0.41	1.11	2.16

Table 1: Spin 0 vs. 0.98 image separation in noise units.

Unknown edge (3-parameter reference pilot). The inner radius was freed to $r_{\text{in}} \in [r_{\text{ISCO}}(a), 8M]$ and marginalized with three priors: conditionally uniform, conditionally log-uniform, and flat in joint (a, r_{in}) , which tilts the spin marginal toward high spin. The pilot used eight truths ($a \in \{0.15, 0.95\} \times i \in \{18^\circ, 78^\circ\} \times$ edge at ISCO or $8M$), each with three noise draws. The original pilot mesh had 2,969 nodes built from 608 new renders. In that preserved pilot, 15/72 prior–observation combinations pass every numerical check, so most numbers are provisional. The following patterns mix accepted and exploratory results and must be read with their acceptance flags (Fig. 5b):

- **Exploratory face-on spin constraints are often weak.** At 18° the free-radius spin s.d. is 0.16–0.29 (mostly ≈ 0.27), against a prior s.d. of 0.28.
- **The ISCO model gives false precision on truncated disks.** In selected case 19, which passed the original finite numerical checks ($a=0.15$, $i=78^\circ$, true edge $8M$), the free-radius 90% interval is $[0.02, 0.73]$. The ISCO-assuming model gives $[0.001, 0.097]$ and *excludes the true spin* with a posterior $6.4\times$ narrower.

- **Spin may survive for high-spin, edge-on disks that reach the ISCO** (s.d. ≈ 0.013 under all priors). The support constraint $r_{\text{in}} \geq r_{\text{ISCO}}(a)$ drives this: a visible edge at $\approx 2M$ is impossible for low spin. These cases failed the original pilot checks; updated flags are reported below.
- **Conditional radius-prior sensitivity is exploratory.** The original all-case median spin-CDF difference between uniform and log-uniform radius is 0.037, including unresolved cases. Flat joint density also changes the spin marginal through prior volume, so it is a different comparison.

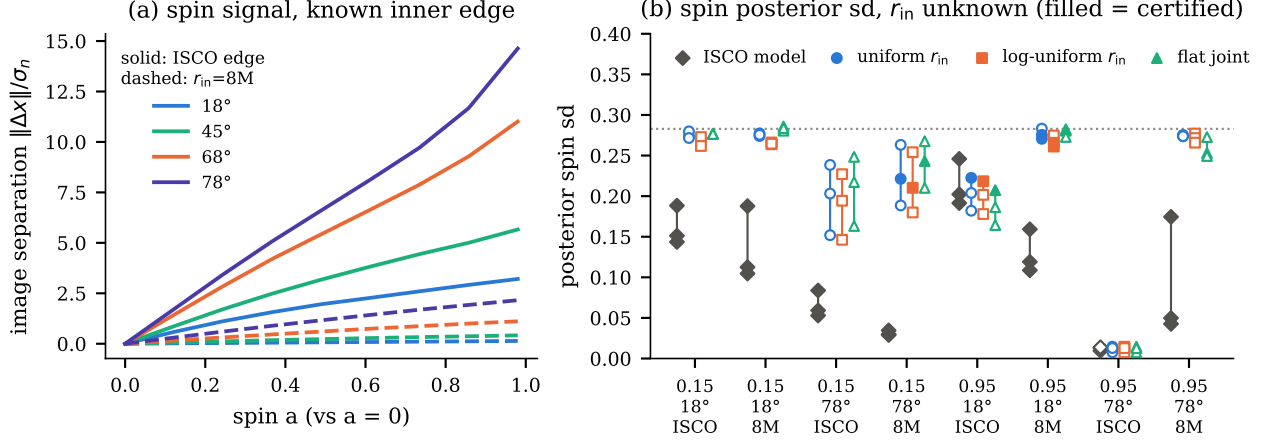


Figure 5: (a) Image separation from $a=0$ as a function of spin, with the disk at the ISCO (solid) or with a known edge at $8M$ (dashed). (b) Original-pilot posterior spin s.d. at the eight pilot truths (labels: spin, inclination, true edge) for the ISCO-only model and three free-radius priors; bars span the three noise draws, filled markers passed the original pilot checks, and the dotted line is the prior s.d. For truths with the edge at $8M$, the ISCO model is misspecified: small widths can be misleading; the selected case-19 example demonstrates exclusion of the true spin, not a population failure rate.

Bounded numerical continuation

Starting from the preserved pilot, two bounded rounds added 128 and 192 nodes, giving a frozen 3,289-node mesh. Selection used measured image errors and included ISCO-face nodes. Slab-based midpoint integration resolved the original quadrature failures on the unchanged mesh: cases 6, 14 and 22 pass $256 \times 256 \times 128$ to $512 \times 512 \times 256$, with largest CDF changes 0.00633, 0.00465 and 0.00200. The remaining cases retain the original quadrature comparison. The priors and numerical thresholds are unchanged.

The final mesh was frozen before drawing seed 2026092603: 96 posterior checks (four per observation), 24 separate ISCO checks and 16 boundary checks. Original truths and boundaries were rechecked too. Fresh posterior image checks pass 83/96; maximum error is 0.899 noise units against 0.25. Two successive forward passes are still required. Final finite acceptance is **0/72 free-radius fits and 24/24 ISCO fits**, with 0 gains and 15 regressions from the original free-radius labels. Of the 72 fits, 35 pass all case-specific checks before the shared boundary gate. The largest original/fresh boundary errors are 0.26011/0.25917, above 0.25; the retained global boundary requirement therefore blocks certification of the whole radius mesh. The changed mesh and expanded fresh-check set both affect this comparison. Adaptive old points are development diagnostics and do not certify the final mesh.

Radius prior	Accepted / 24	All-case median spin s.d.
Conditional uniform radius	0	0.267
Conditional log-uniform radius	0	0.257
Flat joint physical density	0	0.253

Case 19 passes 0/3 free-radius prior fits and passes the ISCO finite checks. It also independently fails a fresh posterior-image check (0.3743 noise units), despite stable forward comparisons. Its uniform-radius spin s.d. is 0.221, versus 0.035 for the misspecified ISCO fit. This remains a selected observation comparison, not a population failure-rate or calibration estimate. Updated high-spin/edge-on ISCO-case acceptance is case 6: 0/3, case 14: 0/3, case 22: 0/3 prior fits. Apparent residual precision is numerically checked only where every criterion passes; the support constraint $r_{\text{in}} \geq r_{\text{ISCO}}(a)$ remains part of its interpretation.

The accompanying local foundation report contains per-case failures, accepted-only versus exploratory summaries, spatial residuals, exact identities and measured costs. No training, reserved-test use, upstream simulator modification or third refinement round occurred. Finite numerical acceptance does not establish continuum ray accuracy or final calibration.

6 What is established, and what is not

Established: an end-to-end, provenance-checked pipeline (simulator wrapper, reusable datasets, JAX trainer, monitor, 225 passing tests); an ISCO-model Fisher map checked at prescribed finite locations and scales; an NPE that learns real image information with near-nominal global coverage; an explicit-likelihood benchmark passing finite numerical checks for 18 selected cases; and strong evidence that the spin constraint in this setup rests largely on the disk-at-ISCO assumption. **Not yet established:** production-scale training (13k images) and SBC on the reserved 1,000-image test set (milestones M5–M6); multi-seed variability; a prior-wide free-radius identifiability map; and any claim about real observations. The emission model, fixed distance and field of view, and Gaussian pixel noise are deliberate idealizations.

7 Suggested directions

The completed two-round numerical study leaves the following later priorities:

1. **Address the remaining numerical limits in a separate bounded study.** Preserve the original radius pilot and its thresholds, re-evaluate accepted cases, and distinguish adaptive development checks from fresh direct renders after the mesh is frozen. Fixed-radius derivative instability leaves Fisher claims unresolved; it does not by itself invalidate posterior refinement. A three-parameter Fisher map is deferred until derivative convergence is demonstrated.
2. **Measure training-seed variability.** Run two additional matched seeds on existing data with the same architecture, optimization schedule, preprocessing and validation rule. Use common development observations and the 18-case reference benchmark. These remain development comparisons, not independent final validation after tuning.
3. **Freeze the M5/M6 protocol before test use.** Record the model-selection rule, reserved seed-2 split identity, posterior-sampling seeds and counts, marginal rank and coverage metrics, uncertainty intervals, and inclination-bin counts. Conditional-on-truth coverage need not be nominal for an exact Bayesian posterior. Marginal checks do not establish joint calibration; a joint diagnostic requires separate implementation and validation. Production training and final calibration remain later compute commitments.